ORACLE VS POSTGRESQL

Auto IDs: Oracle vs PostgreSQL

Sequences and identity columns, side by side. Swipe before your next migration.

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Same idea, different syntax

Both databases hand out unique numbers for your keys.

Both give you two tools: a **sequence** object, or an **identity column**.

The syntax looks close enough to copy. That is exactly where migration scripts break.

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1. Create a sequence

This one works as is in both:

```
CREATE SEQUENCE order_seq
  START WITH 1 INCREMENT BY 1;
```

Each call to get the next value returns a new number: 1, 2, 3...

Tested on PostgreSQL 16. Checked against Oracle documentation

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2. Next value: syntax differs

```
-- Oracle: a pseudocolumn
SELECT order_seq.NEXTVAL
FROM DUAL;

-- PostgreSQL: a function
SELECT nextval('order_seq');
 nextval
---------
       1
```

Tested on PostgreSQL 16. Checked against Oracle documentation

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3. Current value needs NEXTVAL first

```
-- Oracle
SELECT order_seq.CURRVAL FROM DUAL;

-- PostgreSQL (after 2 nextval)
SELECT currval('order_seq');
 currval
---------
       2
```

In both, call NEXTVAL first in your session, or CURRVAL fails.

Tested on PostgreSQL 16. Checked against Oracle documentation

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4. Identity column: same clause

```
-- Oracle 12c+
id NUMBER
  GENERATED ALWAYS AS IDENTITY

-- PostgreSQL 10+
id bigint
  GENERATED ALWAYS AS IDENTITY
```

No separate sequence to create. Insert without the id and the database fills it from its own built-in sequence.

Tested on PostgreSQL 16. Checked against Oracle documentation

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5. ALWAYS rejects your own id

```
INSERT INTO customers (id, name)
VALUES (10, 'Kiran');
```

PostgreSQL:

```
ERROR:  cannot insert a non-DEFAULT
value into column "id"
```

Oracle:

```
ORA-32795: cannot insert into a
generated always identity column
```

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6. PostgreSQL escape hatch

```
INSERT INTO customers (id, name)
OVERRIDING SYSTEM VALUE
VALUES (10, 'Kiran');
INSERT 0 1
```

Or declare the column **GENERATED BY DEFAULT**: then your value wins and the sequence fills the rest.

Tested on PostgreSQL 16

RECAP

Screenshot this

- ✔ CREATE SEQUENCE ... START WITH 1 INCREMENT BY 1 works in both
- ✔ Oracle: seq.NEXTVAL FROM DUAL
- ✔ PostgreSQL: nextval('seq')
- ✔ CURRVAL only after NEXTVAL in your session
- ✔ GENERATED ALWAYS AS IDENTITY: Oracle 12c+, PostgreSQL 10+
- ✔ ALWAYS rejects explicit ids (PostgreSQL: unless OVERRIDING SYSTEM VALUE)

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Migrating between the two?

Send this to a friend who is moving from Oracle to PostgreSQL, or the other way.

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